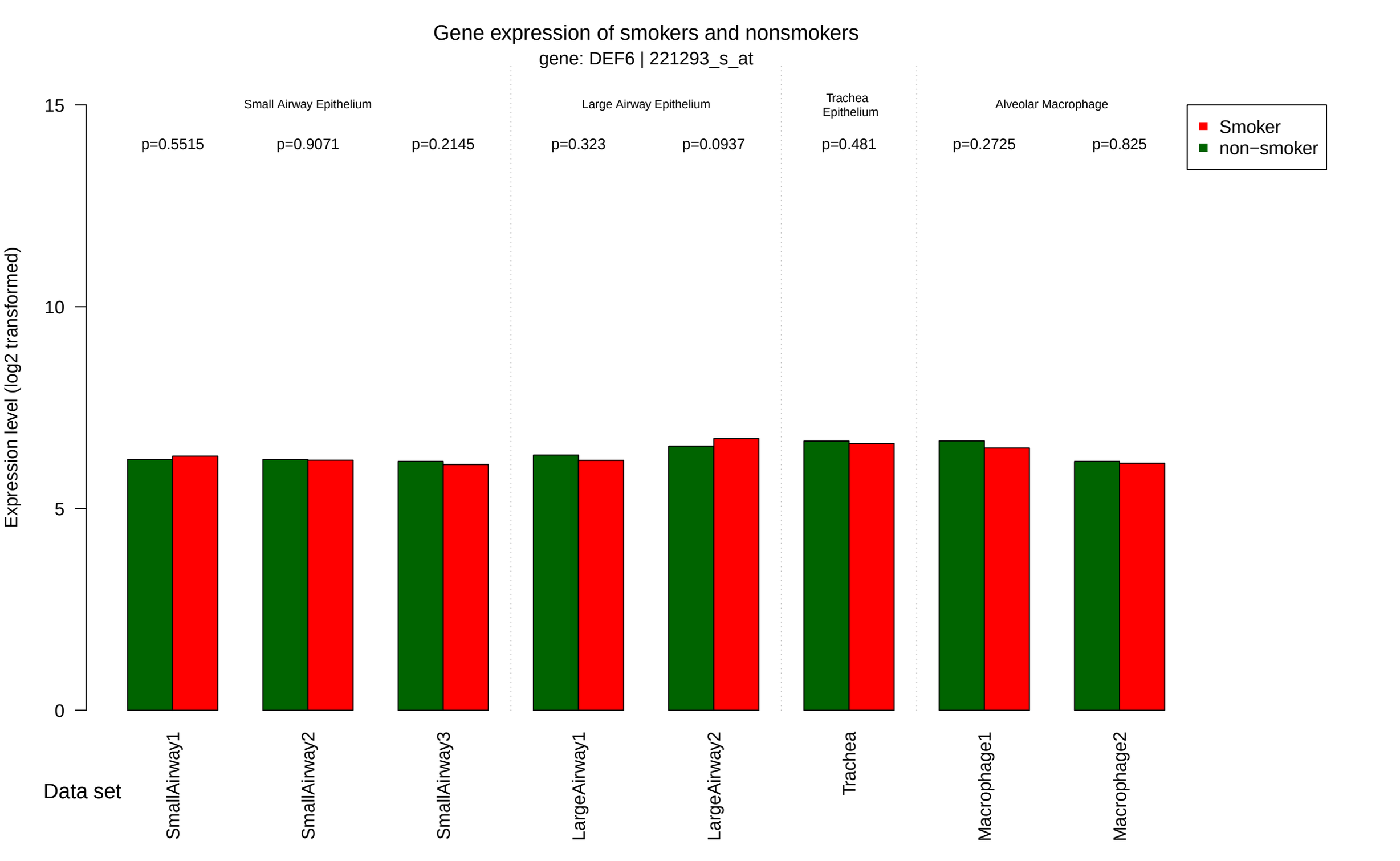
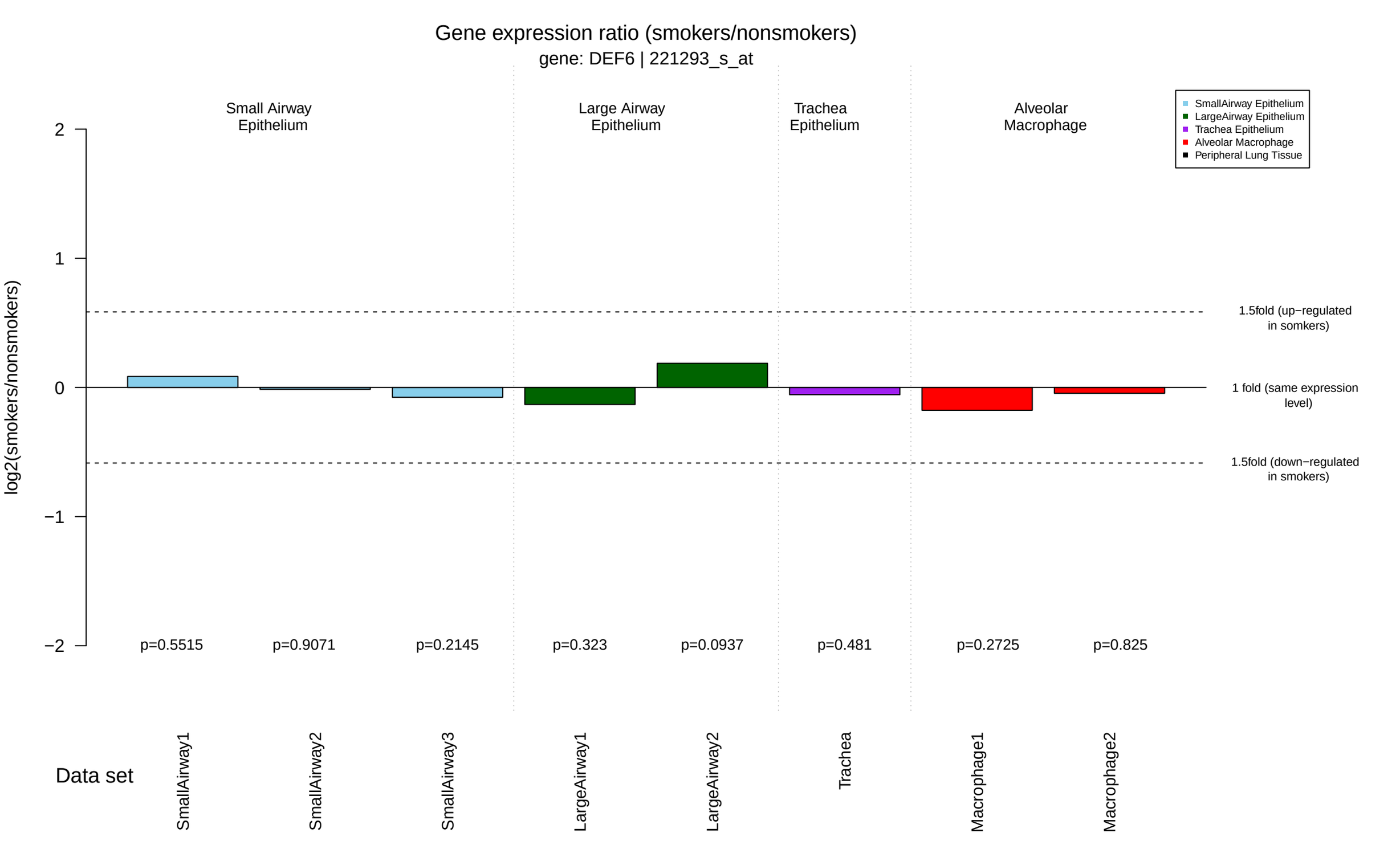
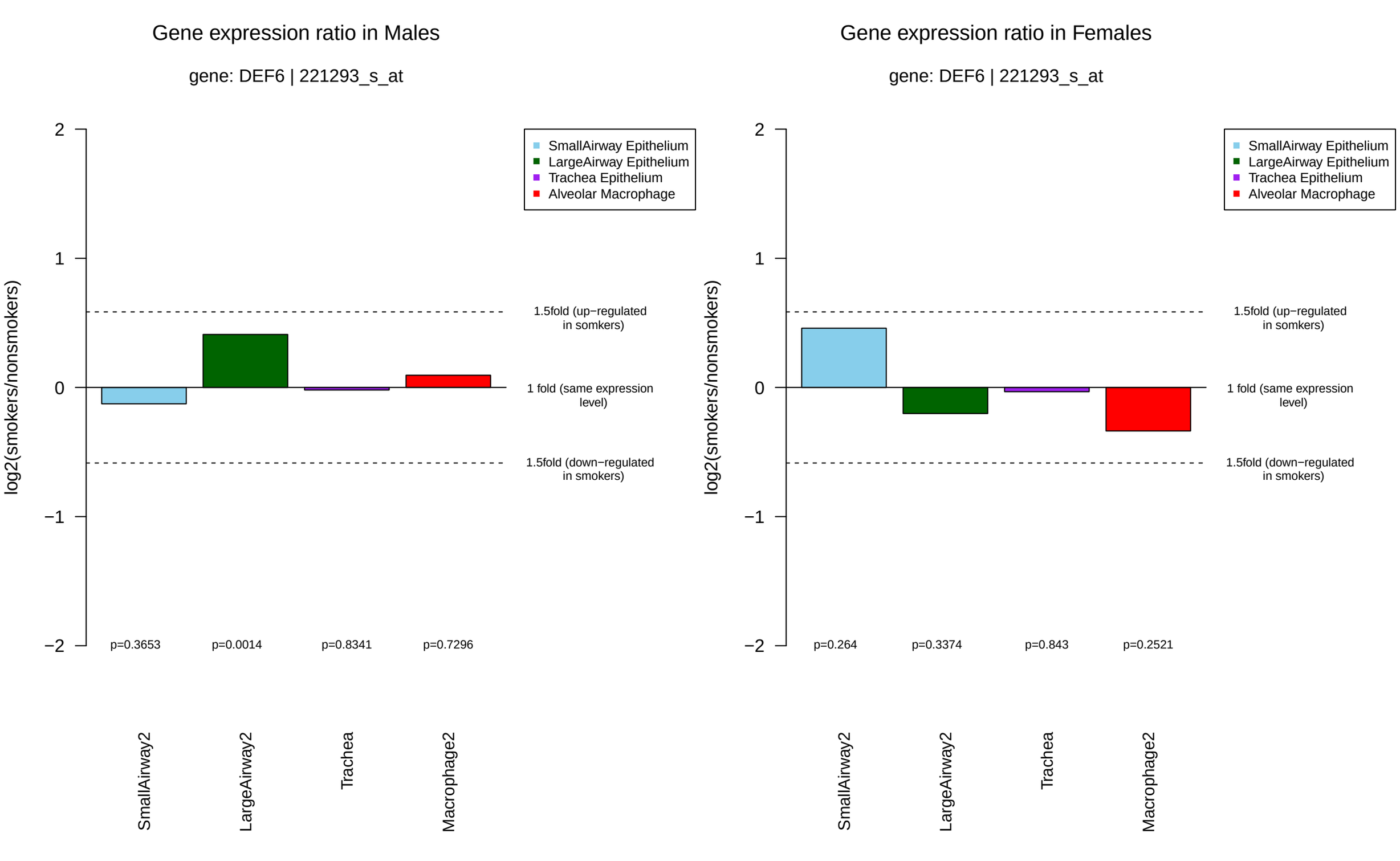
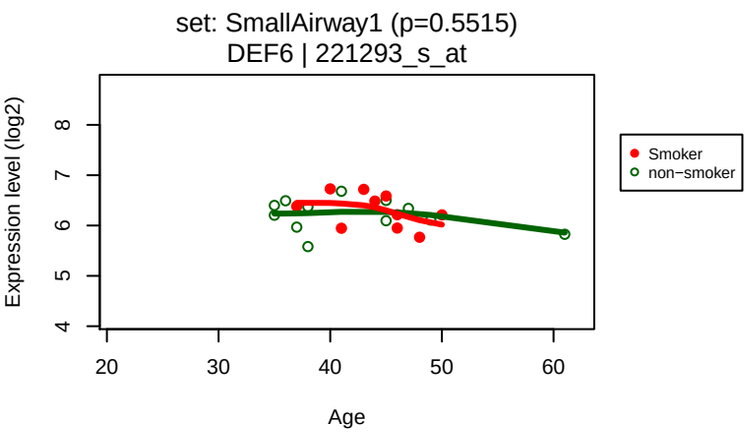
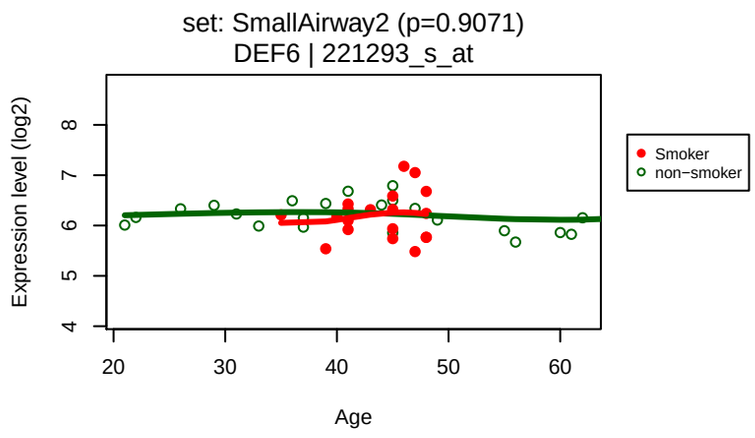
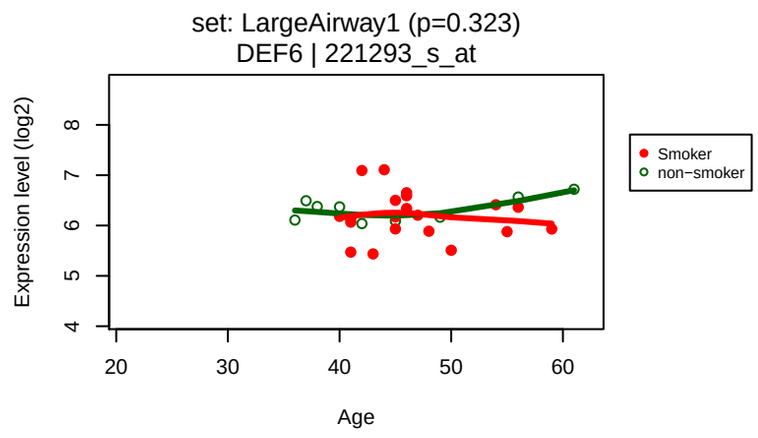

Gene expression of smokers and nonsmokers
gene: DEF6 | 221293_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5515
p=0.9071
p=0.2145
p=0.323
p=0.0937
p=0.481
p=0.2725
p=0.825
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

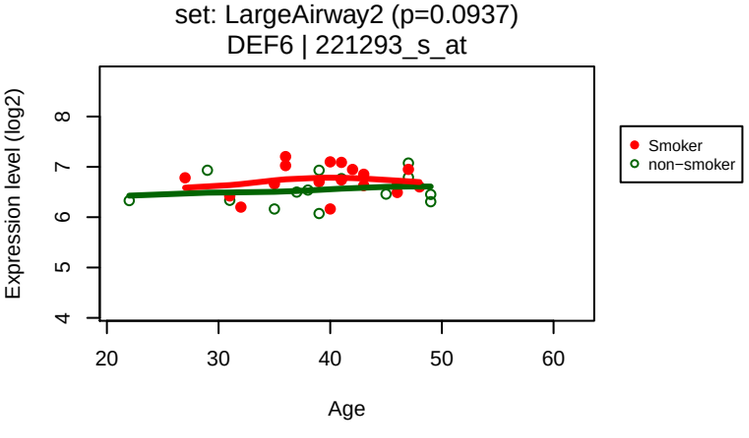

Gene expression ratio (smokers/nonsmokers)
gene: DEF6 | 221293_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5515
p=0.9071
p=0.2145
p=0.323
p=0.0937
p=0.481
p=0.2725
p=0.825
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

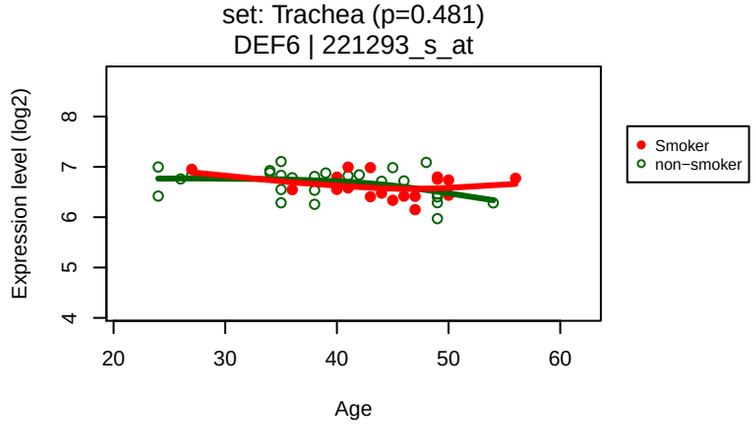

Gene expression ratio in Males
gene: DEF6 | 221293_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3653
p=0.0014
p=0.8341
p=0.7296
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DEF6 | 221293_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.264
p=0.3374
p=0.843
p=0.2521
SmallAirway2
LargeAirway2
Trachea
Macrophage2

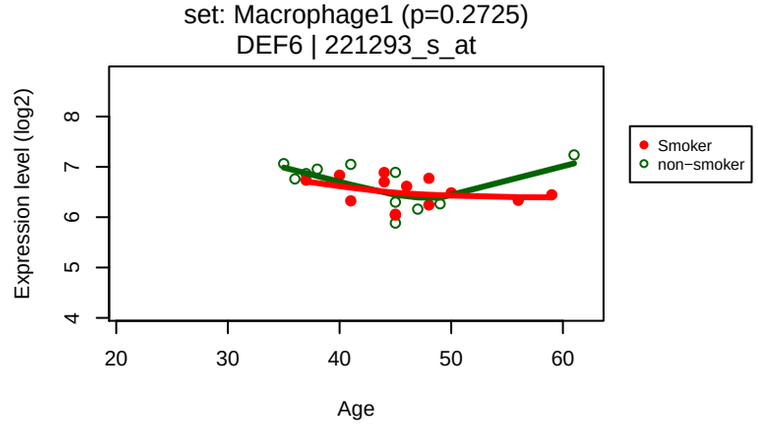

set: SmallAirway1 (p=0.5515)
DEF6 | 221293_s_at
Expression level (log2)
Age
Smoker
non−smoker

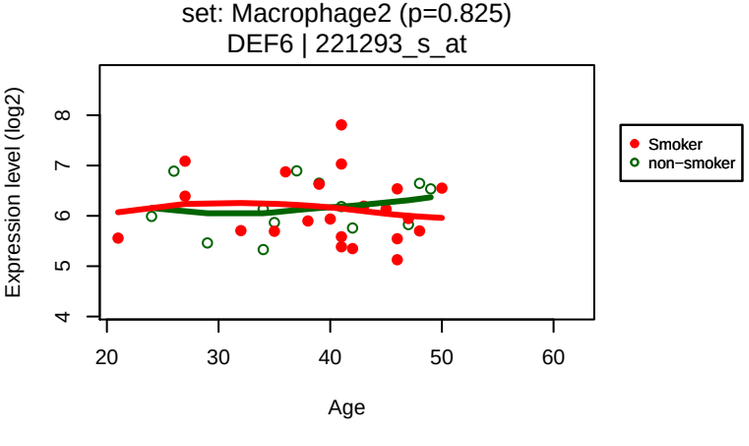

set: SmallAirway2 (p=0.9071)
DEF6 | 221293_s_at
Expression level (log2)
Age
Smoker
non−smoker

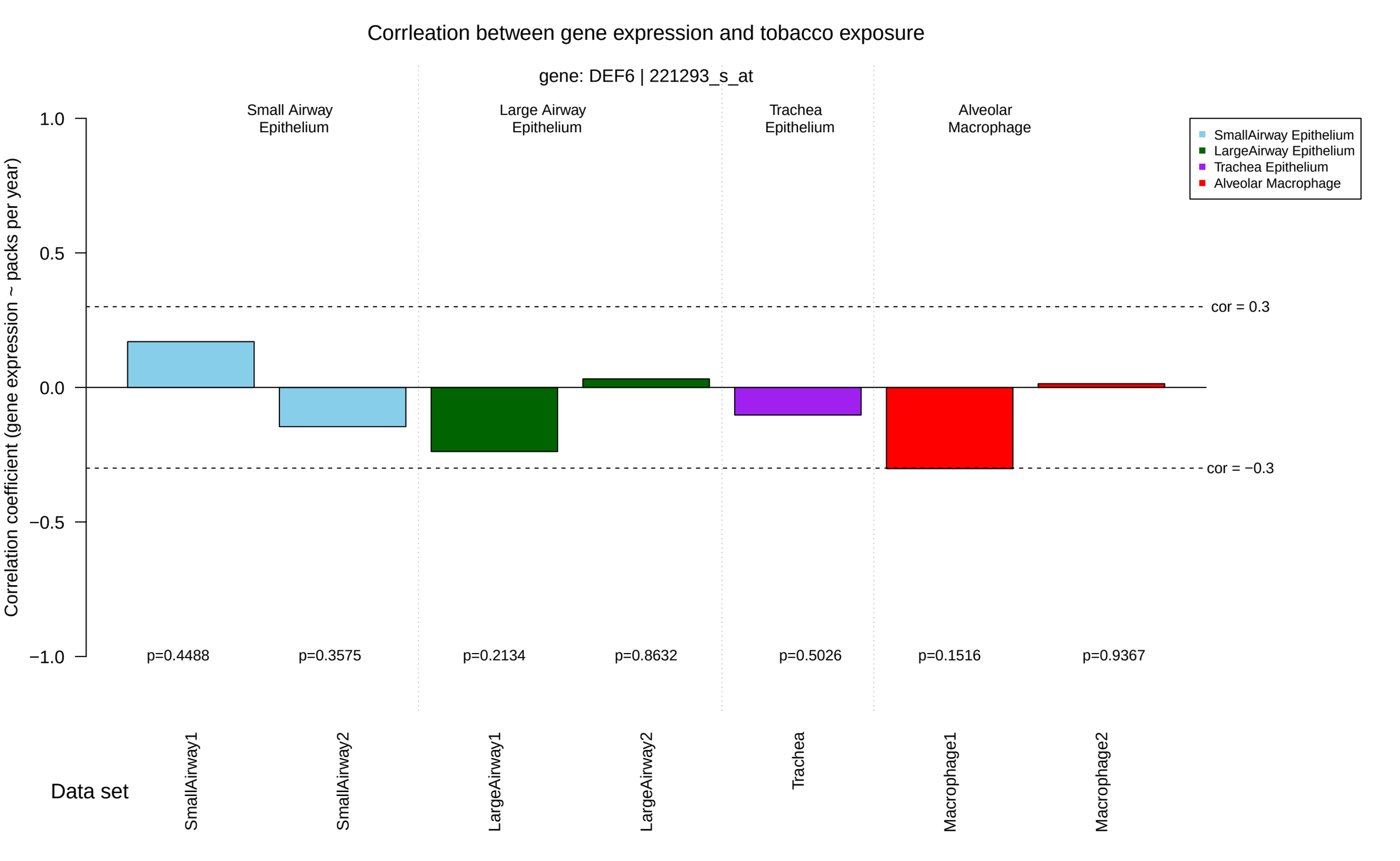

set: LargeAirway1 (p=0.323)
DEF6 | 221293_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0937)
DEF6 | 221293_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.481)
DEF6 | 221293_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2725)
DEF6 | 221293_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.825)
DEF6 | 221293_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DEF6 | 221293_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4488
p=0.3575
p=0.2134
p=0.8632
p=0.5026
p=0.1516
p=0.9367
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2